

Mahesh Prasad Parija, Ph.D.

Research Associate

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EDUCATION

2018	University of Petroleum and Energy Studies, India Ph.D. in Geosciences <i>Thesis: Seismological and Gravity Field Studies on NW Himalaya: Tectonic Implications</i>
2012	Berhampur University, India M.Sc. in Geophysics
2010	Utkal University, India B.Sc. in Physics (Honors)

PROFESSIONAL / RESEARCH EXPERIENCE

2023-Present	Research Associate Virginia Tech, Department of Geosciences
2021-2022	Research Associate National Geophysical Research Institute, Hyderabad, India
2018-2021	Research Associate National Geophysical Research Institute, Hyderabad, India
2013-2018	Research Fellow Wadia Institute of Himalayan Geology, Dehradun, India

RESEARCH INTERESTS

- Earthquake hazard and induced seismicity in stable continental interiors
- Crustal structure and tectonics of active mountain belts, particularly the Himalaya
- Earthquake stress transfer and fault interaction
- Earthquake precursors, including seismic swarms and quiescence
- Machine learning and deep learning methods for seismic monitoring and earthquake source characterization

PEER REVIEWED PUBLICATIONS

10. Ogunleye JB, Conley EW, **Parija MP**, Pollyea RM (2025) "The role of clay layers in managing fluid pressure propagation during managed aquifer recharge", *Groundwater for Sustainable Development*, 31, 101529, doi:10.1016/j.gsd.2025.101529
9. Conley EW, Chambers CR, Ogunleye JB, Koehn LW, Holloway D, Heisig-Mitchell J, Chapman M, **Parija MP**, Pollyea RM (2025) "Pressure management strategies for large-scale aquifer recharge: Mitigating the potential for injection-induced earthquakes", *Journal of Hydrology*, 653, 132767, doi:10.1016/j.jhydrol.2025.132767
8. **Parija MP**, Kumar S, V H A (2023) "Delineation of partial melts and crustal heterogeneities within the crust beneath Kumaon Himalaya, India from Lg wave attenuation", *Scientific Reports (Nature)*, 13, 9922, doi:10.1038/s41598-023-36269-z
7. **Parija MP** (2022) "Earthquake Swarm and Quiescence Precursory to the NE Himalaya 2011 Sikkim Earthquake (M 6.9)", *Journal of the Geological Society of India*, 98, 178–180, doi:10.1007/s12594-022-1955-8
6. Biswal S, Kumar S, Priestley K, Mohanty WK, **Parija MP** (2022) "Structure and Deep Earthquakes beneath the Kinnaur, NW Himalaya: Constraints from Relocated Seismicity and Moment Tensor Analysis", *Lithosphere (GSA)*, 2022(6), 2834594, doi:10.2113/2022/2834594
5. **Parija MP**, Kumar S, Tiwari VM, Biswal S, Biswas A, Velliyidathu A (2021) "Coulomb stress modeling and seismicity in the western Himalaya, India since 1905: Implications for the incomplete ruptures of the Main Himalayan Thrust", *Tectonics*, 40, 1–16, doi:10.1029/2020TC006204
4. Biswal S, Kumar S, Roy SK, Kumar MR, Mohanty WK, **Parija MP**, Paul A (2020) "Upper Mantle Anisotropy beneath the Western Segment, NW Indian Himalaya, Using Shear Wave Splitting", *Lithosphere (GSA)*, 1–9, doi:10.2113/2020/8856812

3. **Parija MP**, Kumar S, Tiwari VM, Purnachandra Rao N, Kumar N, Biswal S, Singh I (2018) "Microseismicity, tectonics and seismic potential in the Western Himalayan segment, NW Himalaya, India", *Journal of Asian Earth Sciences*, 159, 1–16, doi:10.1016/j.jseaes.2018.03.016
2. Biswas A, **Parija MP**, Kumar S (2017) "Global nonlinear optimization for the interpretation of source parameters from total gradient of gravity and magnetic anomalies caused by thin dyke", *Annals of Geophysics*, 60(2), 1–17, doi:10.4401/ag-7129
1. **Parija MP**, Kumar S, Biswal S, et al. (2016) "A preliminary one-dimensional crustal velocity model for Himachal Pradesh, India", *Journal of Seismology*, 20, 305–318, doi:10.1007/s10950-015-9528-6

BOOK CHAPTERS

2. **Parija MP**, Biswal S, Biswas AK (2021) "Static stress change from February 6, 2017 (M 5.8) earthquake, Northwestern Himalaya, India", in *Basics of Computational Geophysics* (Samui, Dixon, Bui, eds.), Elsevier, pp. 213–222, doi:10.1016/B978-0-12-820513-6.00020-5
1. Trivedi S, Kumar P, **Parija MP**, Biswas A (2020) "Global Optimization of Model Parameters from the 2-D Analytic Signal of Gravity and Magnetic Anomalies Over Geo-Bodies with Idealized Structure", in *Advances in Modeling and Interpretation in Near Surface Geophysics* (Biswas & Sharma, eds.), Springer, Cham, doi:10.1007/978-3-030-28909-6_8

MANUSCRIPTS IN PREPARATION / SUBMITTED

2. **Parija MP**, Chapman MC, Biswal S, Acosta M, Conley EW, Ogunleye JB, Pollyea RM "Microseismic investigations of Intraplate Earthquakes: A case study from Hampton Roads, Southeastern Virginia, USA", in preparation
1. Conley EW, Chapman MC, **Parija MP**, Ogunleye JB, Pollyea RM "Assessing Induced Earthquake Detection for Industrial-Scale Fluid Injection Operations", in revision at *Seismological Research Letters*

RESEARCH FUNDING

4. USGS: *Developing a High-Resolution Earthquake Catalog and Imaging Active Fault Structures in the Giles County Seismic Zone Using a Temporary STRYDE Node Network* (2026: Total \$86,505; Bemis PI, **Parija Co-PI**) [Pending]
3. NSF: *Natural and Induced Earthquake Classification using Input- and Output-Constrained Physics-Informed Neural Networks* (2026–2028: Total \$347,858; **Parija Co-PI**) [Pending]
2. DOE: *Central Appalachian Regional Depot for Industrial CO₂ Abatement in Lone Star, Virginia (Project CARDINAL)* (2024–2027: Total \$9,000,000; **Parija, Senior Personnel**) [Pending]
1. HRSD–SWIFT: *Seismic monitoring & earthquake hazard assessment for managed aquifer recharge operations in southeast Virginia* (2021–2026: Total \$1,451,052; **Parija, Research Associate/Senior Personnel**) [Active]

FELLOWSHIPS & HONORS

- Postdoctoral Research Associateship, National Academy of Sciences, India, 2021
- Institute Postdoctoral Research Associateship, CSIR–National Geophysical Research Institute, Hyderabad, India, 2018
- Senior Research Fellowship, Wadia Institute of Himalayan Geology, India, 2015
- Junior Research Fellowship, Wadia Institute of Himalayan Geology, India, 2013
- International Travel Grant, SERB, Department of Science and Technology, Govt. of India, 2022
- Student Travel Grant Award, American Geophysical Union, Washington D.C., USA, 2018

INVITED LECTURES & PRESENTATIONS

8. Invited Speaker, Hampton Roads Sanitation District–SWIFT Workshop, Nansemond, VA (June 27, 2024)
7. Invited Speaker, SWIFT Research Center, Nansemond, VA (June 6, 2023)
6. Speaker, Joint Scientific Assembly IAGA–IASPEI, Hyderabad, India (2021)
5. Speaker, American Geophysical Union Fall Meeting, Washington, D.C., USA (2018)

4. Invited Speaker, Vietnam Institute for Advanced Studies in Mathematics (VIASM), Hanoi, Vietnam (December 2017)
3. Speaker, Workshop on Earthquake and Tsunami: Modeling and Observations, International Center for Theoretical Physics (ICTP), Hanoi, Vietnam (2017)
2. Speaker, National Geo-Research Scholar's Meeting, Dehradun, India (2016)
1. Speaker, National Workshop on the Status of Natural Hazards in Himachal Pradesh, Dharamshala, India (2014)

CONFERENCE ABSTRACTS

9. **Parija MP**, Chapman MC, Biswal S, Conley EW, Ogunleye J, Pollyea RM (2025) "Shear Wave Velocity Structure and Moho Depth Beneath the Virginia Coastal Plain (VCP) From Fundamental-mode Rayleigh-wave Group-velocity", Abstract 134, Seismological Society of America Annual Meeting, Baltimore, MD
8. **Parija MP**, Chapman MC, Pollyea RM (2023) "Statistical analysis of the foreshock-mainshock-aftershock sequence: a case study of 2020 Mw 5.1 Sparta, North Carolina, earthquake", AGU Fall Meeting, S21F-0353
7. **Parija MP**, Gupta HK (2022) "Analysis of the 2015 Gorkha-Dolakha (Central Nepal) Foreshocks and Aftershocks Sequence Through Transients in B Values", *Seismological Research Letters*, 93(2B), 1115–1372
6. Kumar S, Sain K, **Parija MP**, et al. (2021) "Re-Appraisal of Seismicity and Seismotectonics of Himalaya, India: A Review of Historical and Current Status", *Journal of the Indian Geological Congress*, 2(2) and 13(1&2), 45–59
5. Biswal S, Kumar S, Mohanty W, **Parija MP** (2019) "Imaging the lithospheric structure across NW Himalaya, India utilizing local seismic tomography", *Geophysical Research Abstracts*, 21
4. Kumar S, **Parija MP**, Biswal S (2018) "Active Tectonics and Potential of Seismic Hazard in the Northwest Himalayan Region, India", AGU Fall Meeting, S14B-08
3. **Parija MP**, Kumar S, Tiwari VM (2018) "Crustal Structure, Stress Field Evolution across NW Himalaya, India from a Combined Seismic and Gravity Study", AGU Fall Meeting, T22A-08
2. Kumar S, Biswal S, **Parija MP** (2016) "Active Tectonics of Himalayan Faults/Thrusts System in Northern India on the basis of Recent & Paleo Earthquake Studies", AGU Fall Meeting, S44B-06
1. Kumar S, **Parija MP**, Biswal S (2016) "Microseismicity, Tectonics and Seismic Potential in the Western Himalayan Segment, NW Himalaya (India) Region", AGU Fall Meeting, S53A-2830

FIELD EXPERIENCE

- Hampton Roads Seismic Network — operate a five-station real-time broadband seismic network monitoring southeastern Virginia (2023–present)
- Giles County Seismic Zone STRYDE Nodal Deployment — co-led a 10-station dense nodal seismic deployment to monitor and locate microseismicity in an intraplate setting (2026)
- Northwest Himalaya Broadband Network — installation, operation, and maintenance of a 32-station broadband seismic network in the high-altitude NW Himalaya (2018–2021)

SUPERVISORY ROLES

Graduate Students

- Ethan W. Conley (Virginia Tech: mentored Fall 2024). *Now: Postdoctoral Fellow, Sandia National Laboratories*
- John B. Ogunleye (Virginia Tech: mentored Fall 2024–Spring 2026). *Now: Hydrogeologist, Kansas State Geological Survey*

PROFESSIONAL SERVICE

- Guest Editor, Advances in Statistical and Computational Seismology Collection, Scientific Reports (Nature), 2024–Present
- Editorial Board Member, Scientific Reports (Springer Nature), 2023–Present
- Editorial Board Member, Scientific Reviews (Springer Nature), 2026–Present
- Reviewer, Journal of Geophysical Research – Solid Earth (AGU), 2024–Present

- Reviewer, Natural Hazards (Springer Nature), 2024–Present
- Reviewer, International Journal of Geophysics (Hindawi), 2023–Present
- Reviewer, Journal of Earth System Sciences (Springer Nature), 2021–Present
- Reviewer, Scientific Reports (Springer Nature), 2023–Present